

Random Forest



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Supervised Learning

Supervised Learning

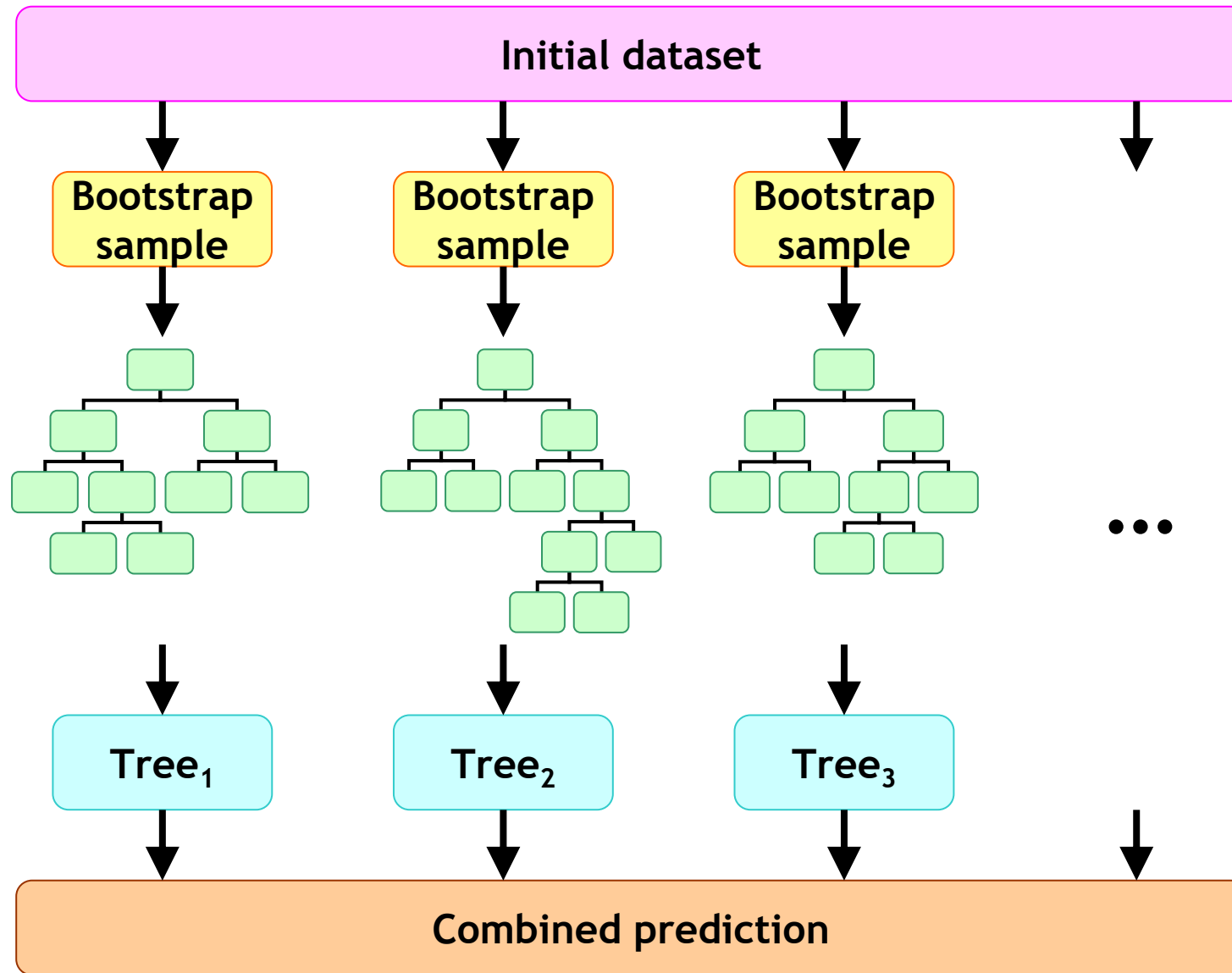
[Random Forest]

Random Forest -concept

Definition

- **Random forest** (or **random forests**) is an ensemble classifier that consists of many decision trees and outputs the class that is the mode of the class's output by individual trees.
- The term came from **random decision forests** that was first proposed by Tin Kam Ho of Bell Labs in 1995.
- The method combines Breiman's "bagging" idea and the random selection of features.

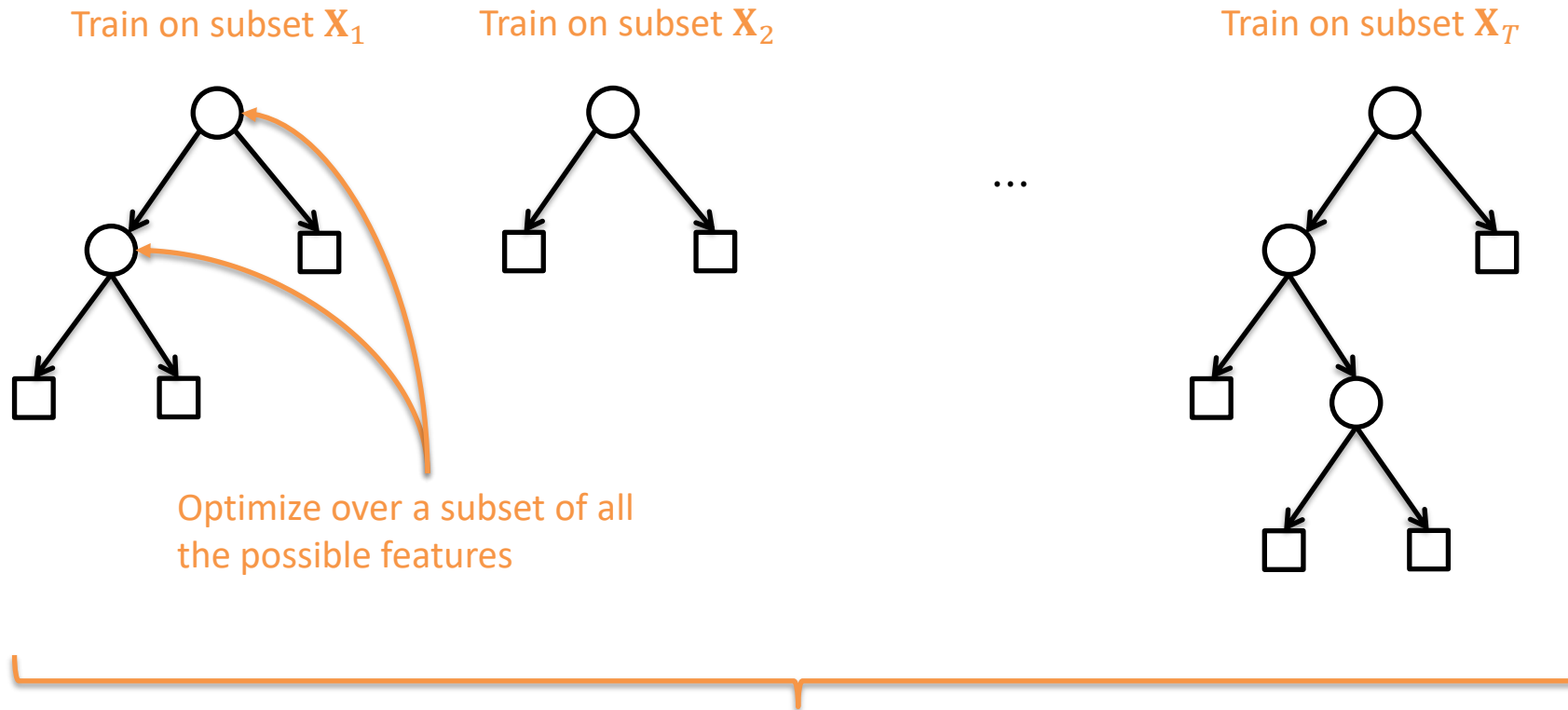
Random Forest -concept



Random Forest -concept

Random forest:

Ensemble of T decision trees



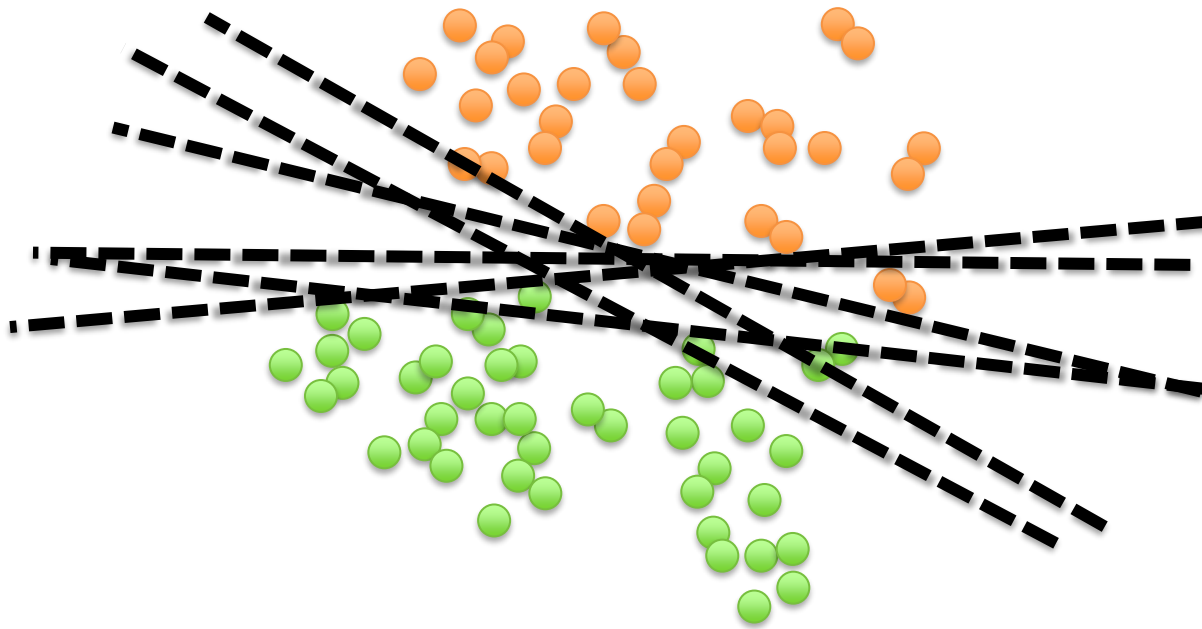
Define an ensemble decision rule, e.g. $p(c|x, T) = \frac{1}{T} \sum_{i=1}^T p(c|x, T_i)$

Random Forest -concept

Decision forests:

Max-margin behaviour

$$p(c|\mathbf{x}, T) = \frac{1}{T} \sum_{i=1}^T p(c|\mathbf{x}, T_i)$$

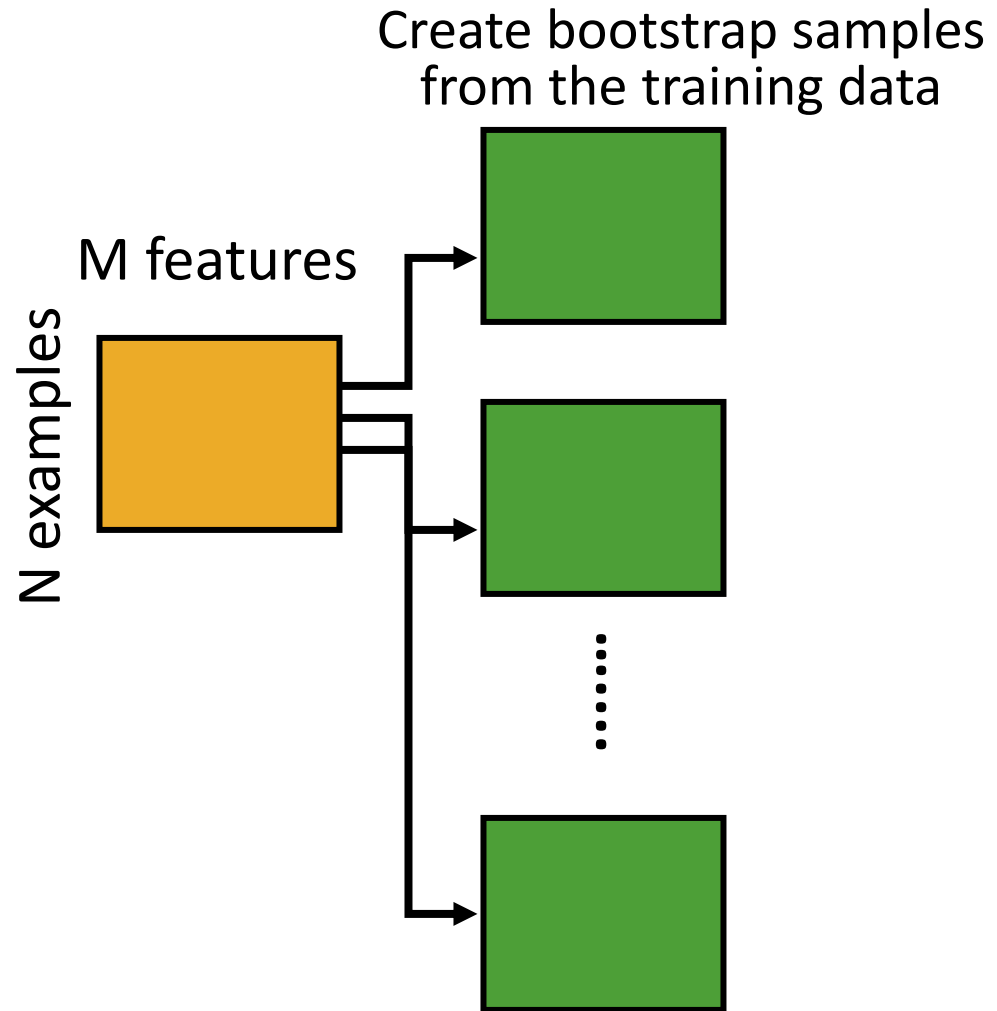


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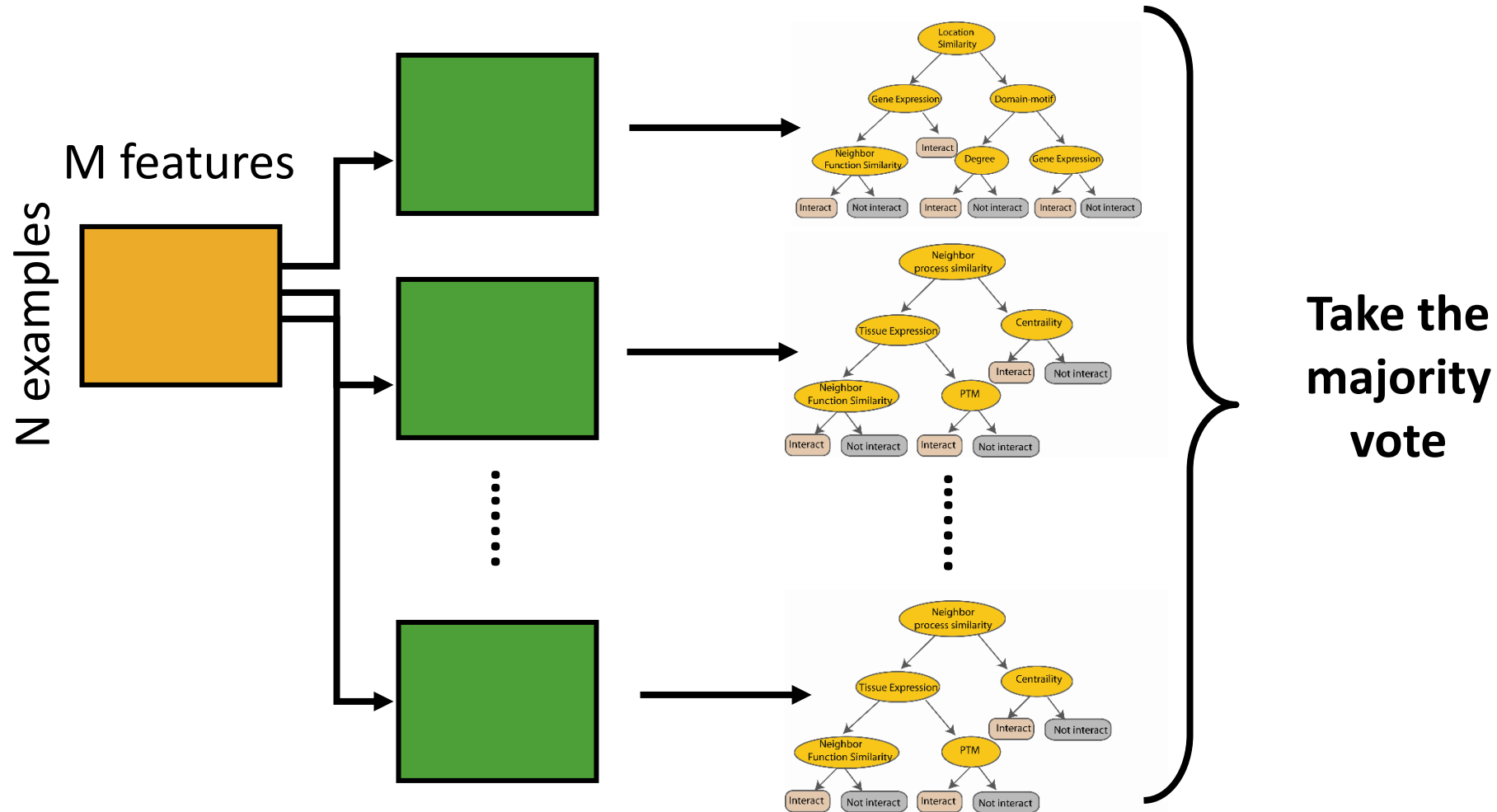
Random Forest advantages:

1. RF models are robust to over-fitting.
2. There is no need in pre-selection of variables.
3. RF has its own reliable procedure for estimation of predictive ability of model.
4. RF models are robust to “noise” in training dataset.
5. RF allows to estimate variable importance for target property (interpretability of RF model).
6. RF allows to analyze compounds with different mechanisms of action.
7. RF method is very fast and effective in working with huge datasets.

Bagging

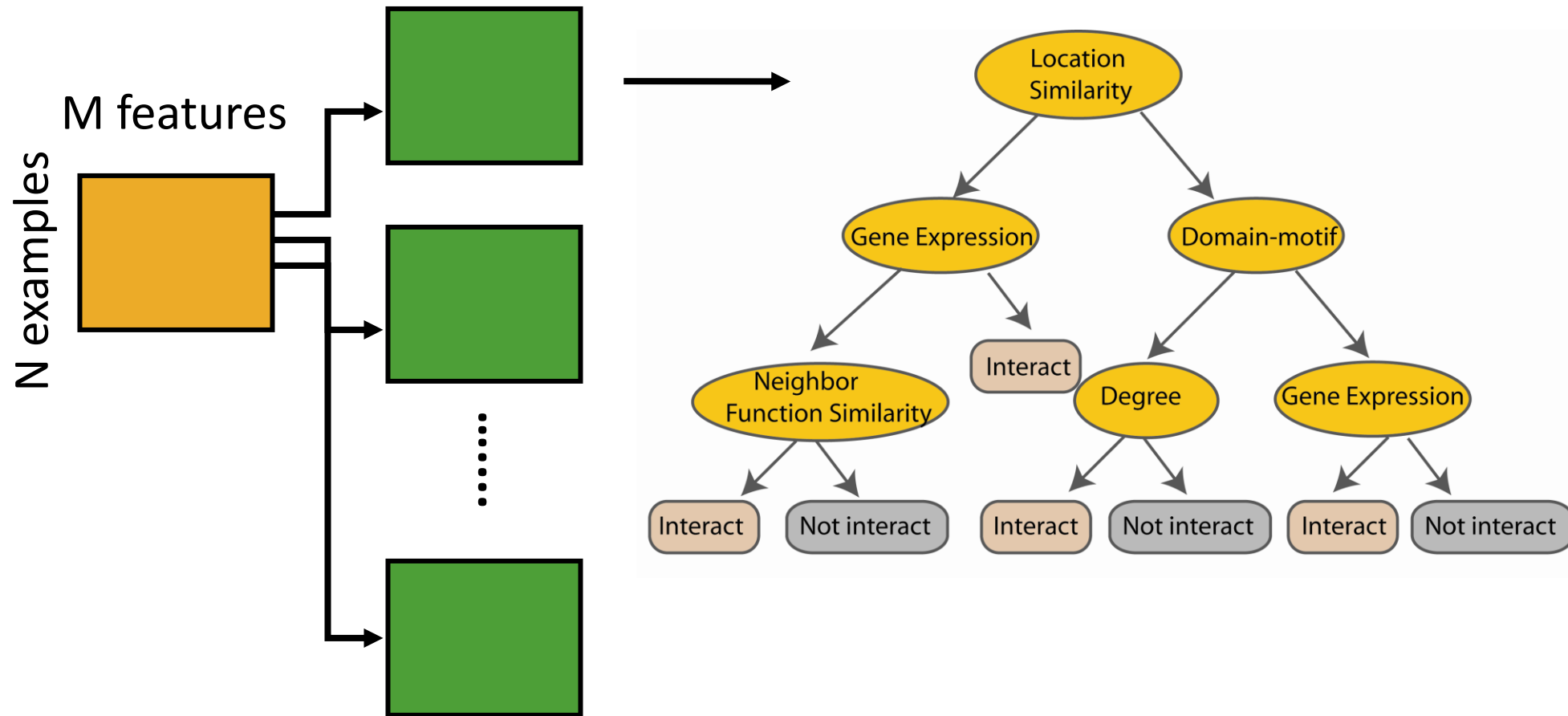


Random Forest Classifier



Random Forest Classifier

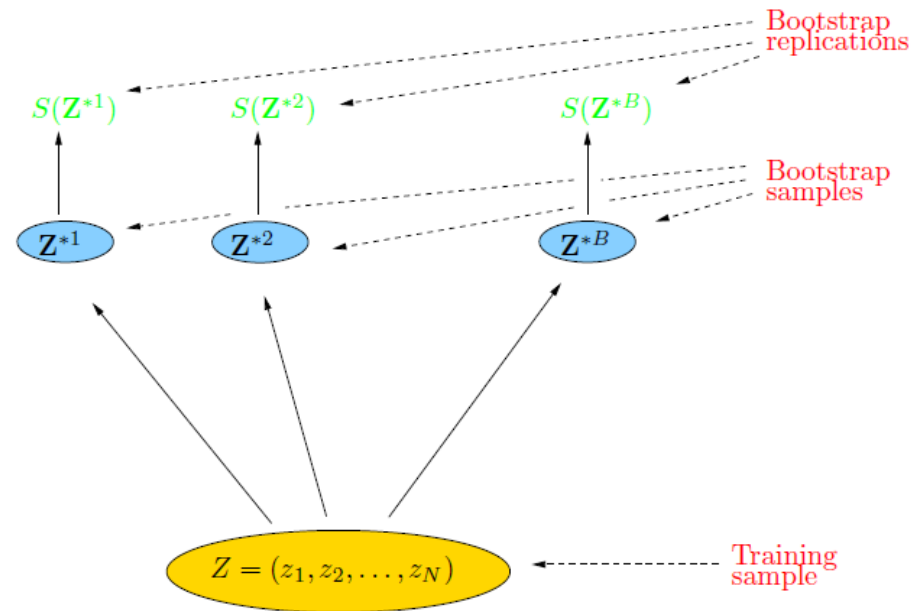
Construct a decision tree



Bootstrap

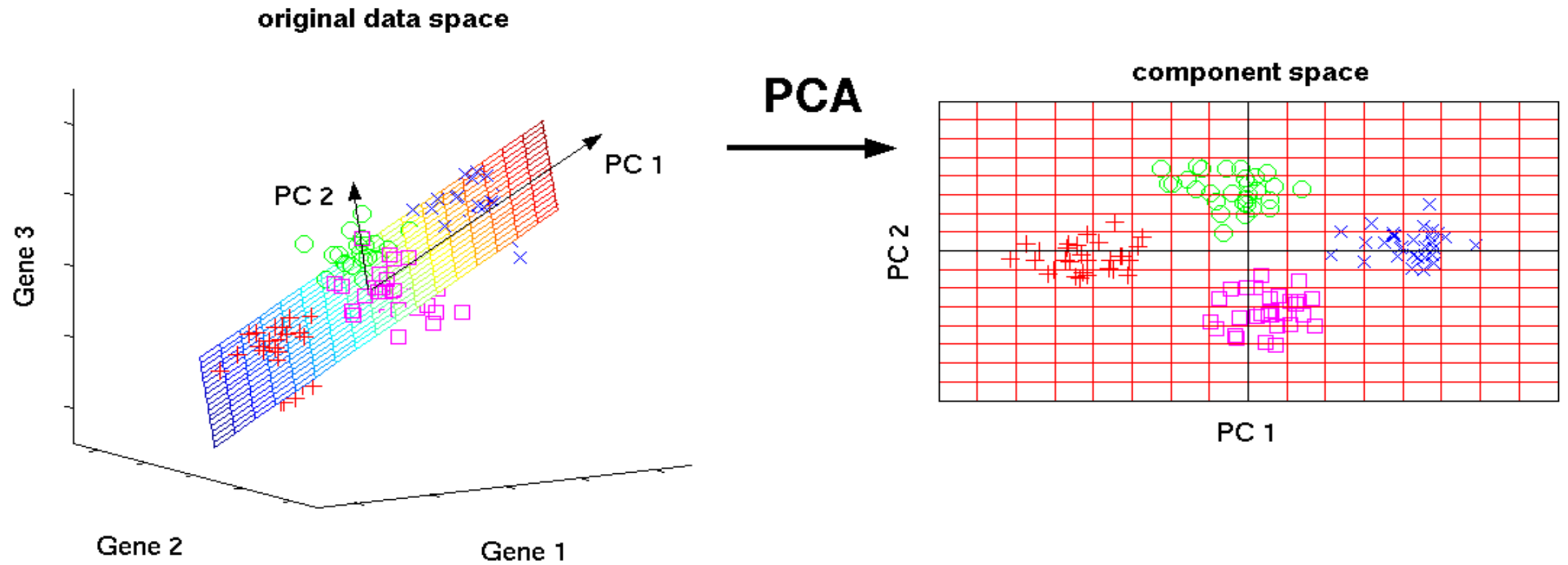
The basic idea:

randomly draw datasets *with replacement* from the training data, each sample *the same size as the original training set*

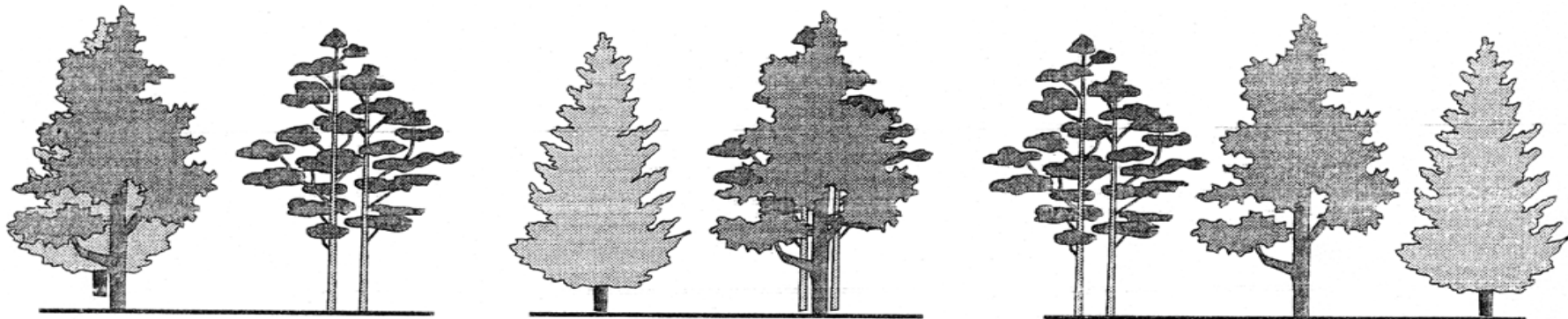
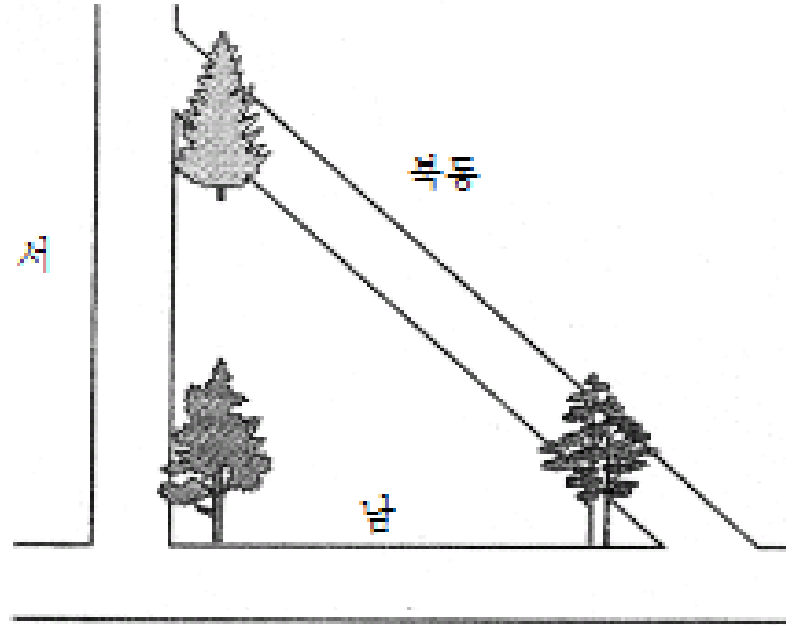


Curse of Dimensionality

Dimensionality Reduction



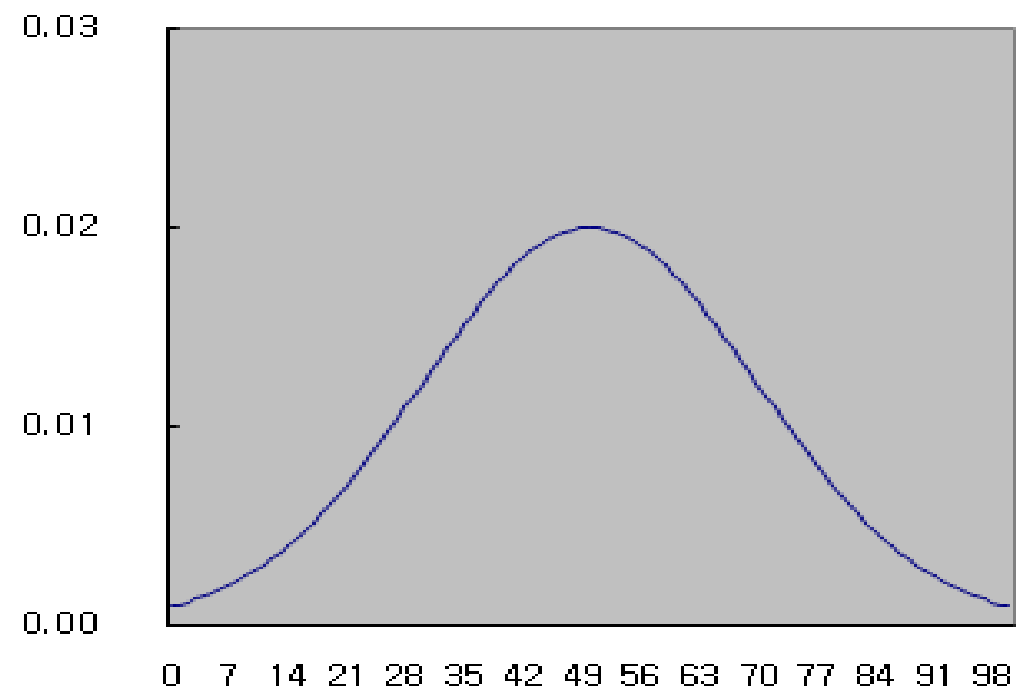
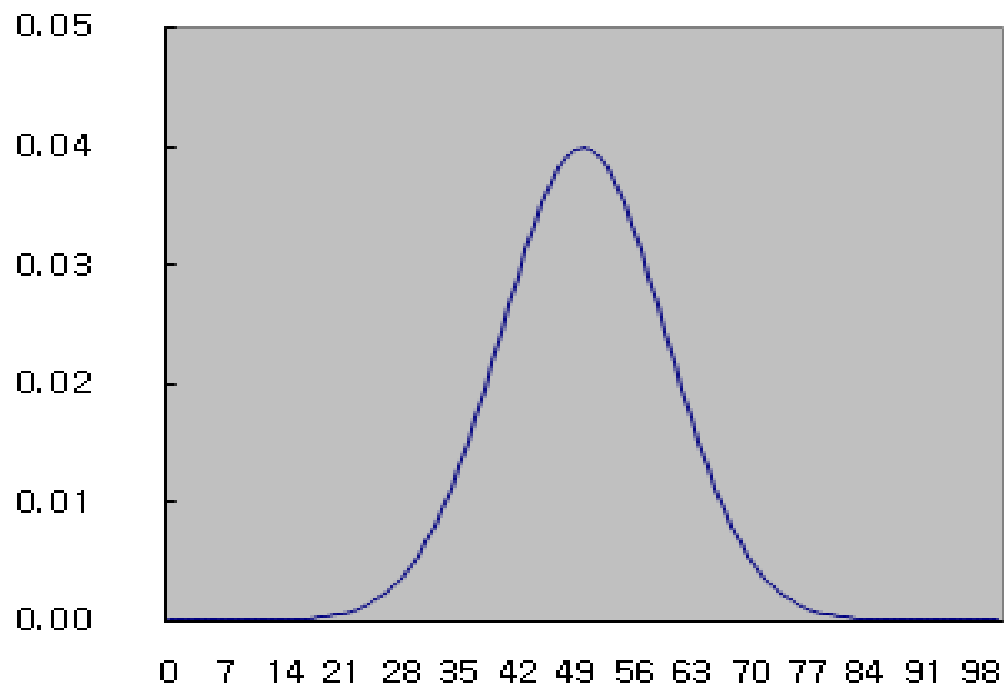
PCA



Name	Design	Functionality
V	40	90
A	90	40
C	70	70

Name	Design	Functionality	Design + Functionality	Design - Functionality
V	40	90	130	-50
A	90	40	130	50
C	70	70	140	0

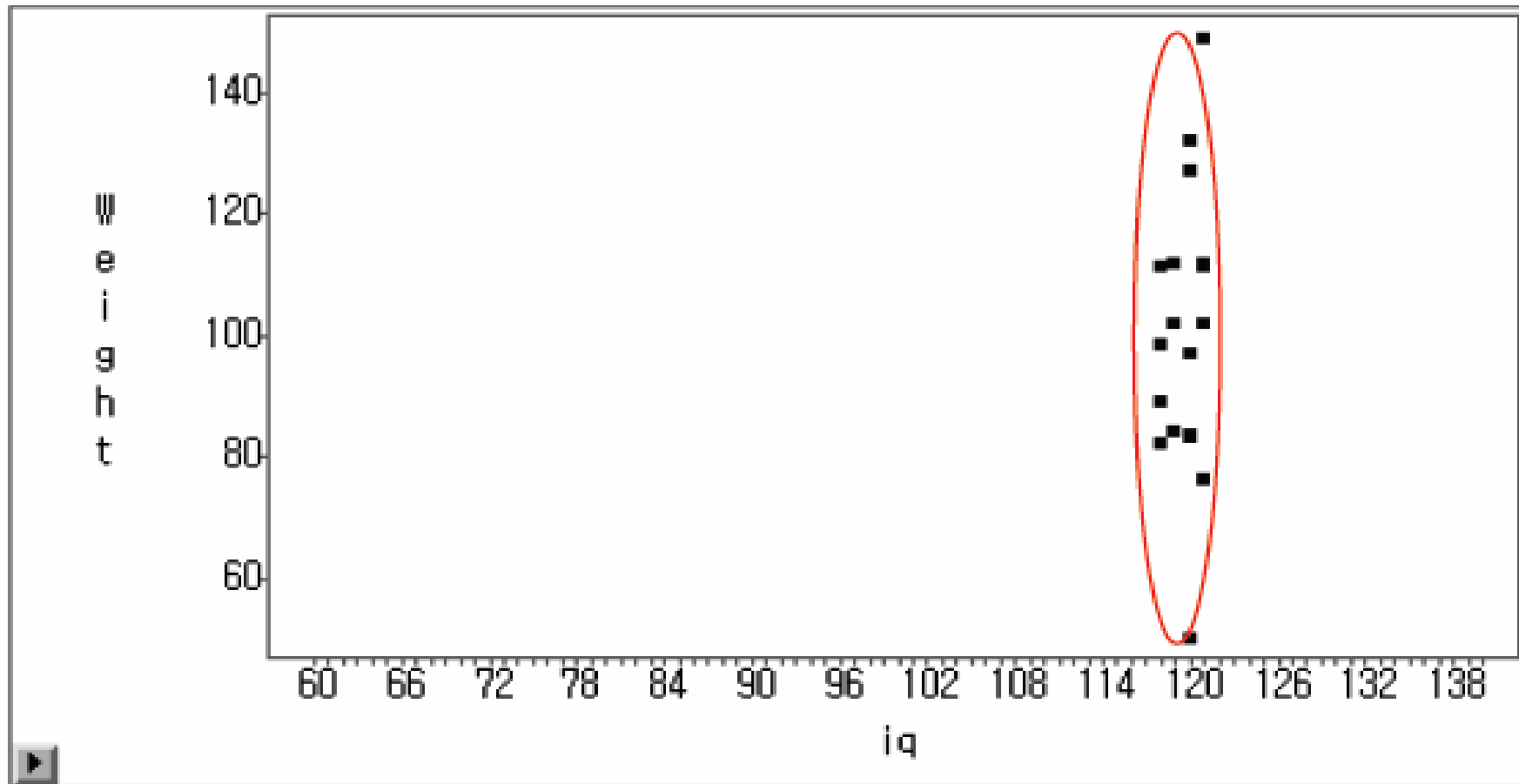
PCA



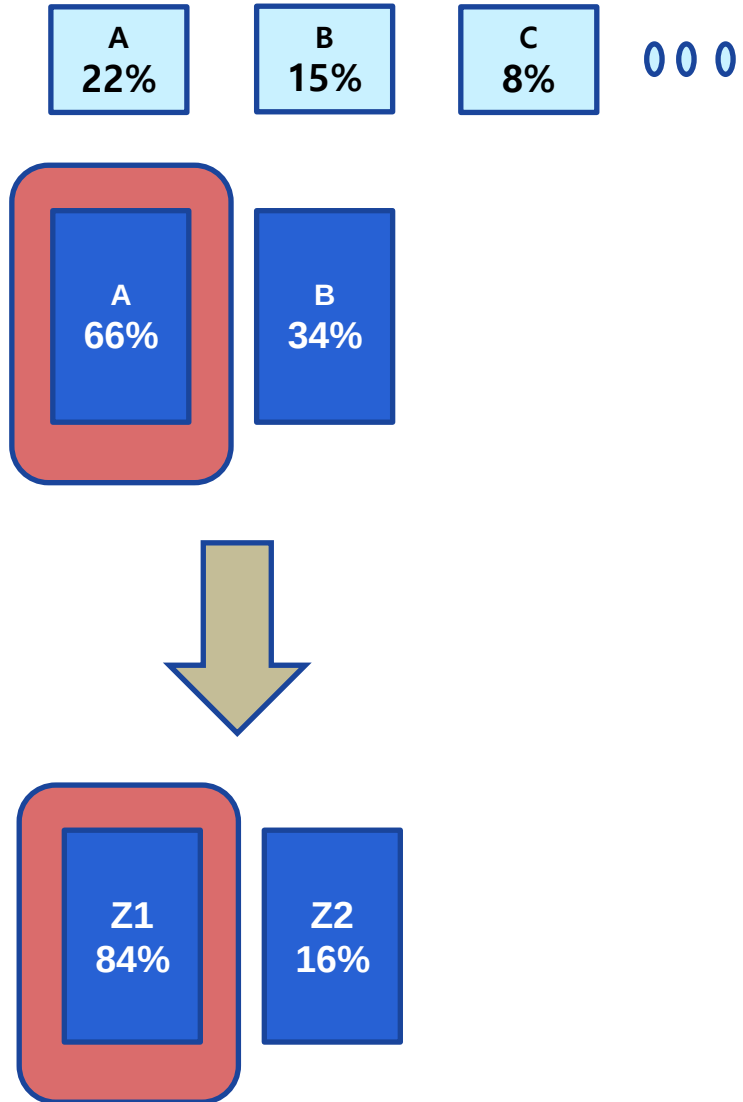
PCA

Data	Design + Functionality	Design - Functionality
Variance	33.3	2500

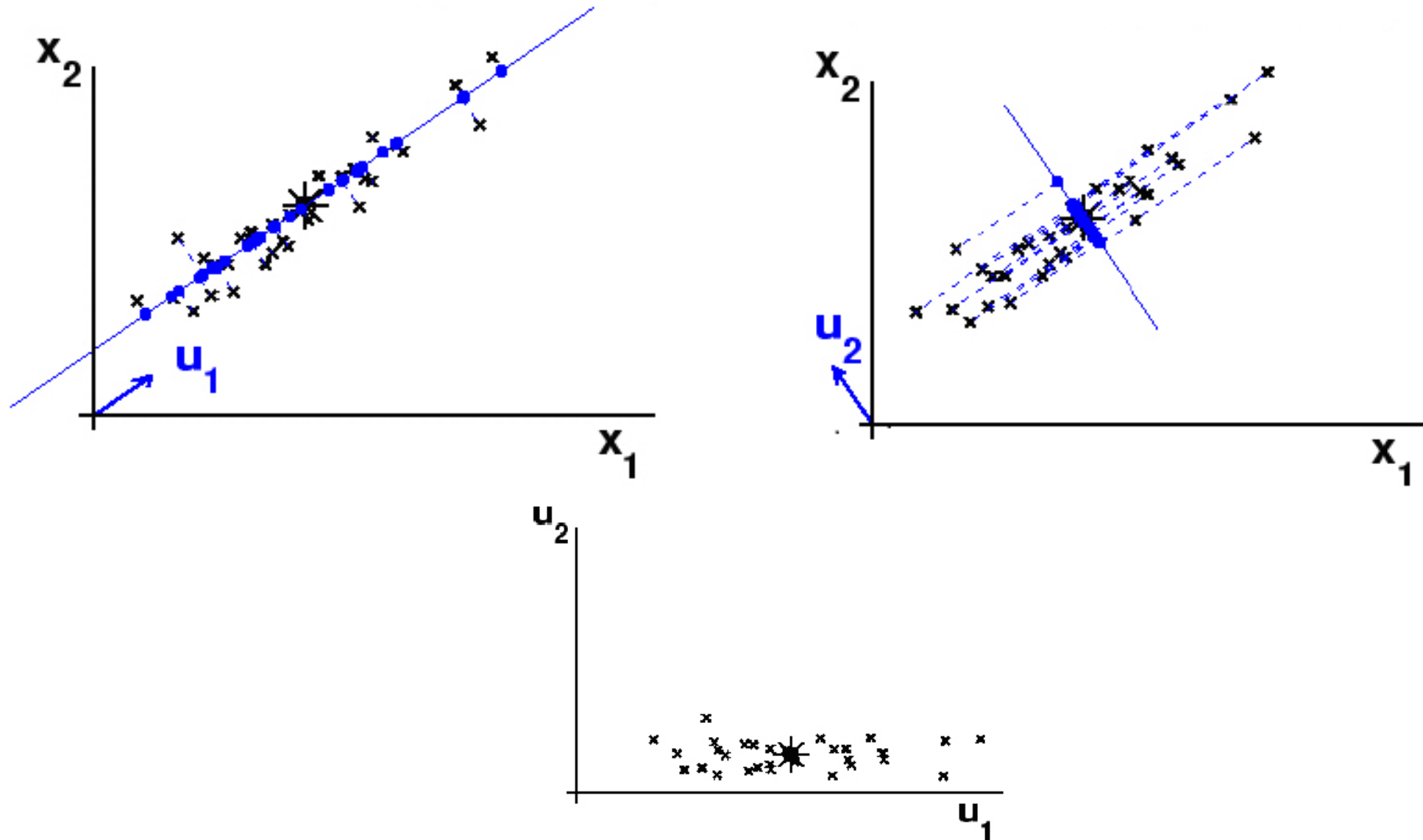
PCA



PCA



PCA



Question and Answer
THANKYOU 😊